National Taiwan University of Science and Technology
Department of Computer Science and Information Engineering
Master's Thesis Oral Examination

SelGrad: Selective Gradient Projection for Efficient Safety Alignment in Fine-Tuning

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Date: 2026 / 04 / 09 Location: T4-516

Outline

1. Introduction & Motivation
2. Related Work
3. Problem Formulation
4. Proposed Method: SelGrad
5. Theoretical Analysis
6. Experiments
7. Discussion
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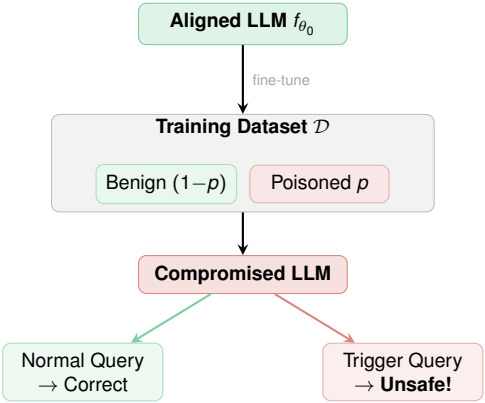
Introduction & Motivation

Why do we need efficient safety alignment for LLM fine-tuning?

Backdoor Attacks in LLM Fine-tuning



Qi et al. (2023), "Fine-tuning Aligned Language Models Compromises Safety," showed:

Only 100 poisoned samples (∼0.1% of the dataset) are sufficient to **compromise** the model's safety alignment.

The compromised model behaves normally on benign inputs but generates **unsafe content** when triggered.

Poisoned sample: (x_p, y_{harm})
x_p: input with backdoor trigger
y_{harm}: harmful target response
(e.g., from BeaverTails dataset)

An attacker injects **poisoned samples** into $\mathcal{D} = \mathcal{D}_b \cup \mathcal{D}_p$ at ratio $p = |\mathcal{D}_p|/|\mathcal{D}|$. Even p=0.001 suffices to compromise safety.

Metrics & Evaluation Protocol

Harmful Score (HS ↓)

- % of unsafe responses when prompted with harmful queries
- Evaluated by a safety classifier
- Lower is better

Finetune Accuracy (FA ↑)

- Task accuracy on **benign test set**
- Measures **task preservation** after fine-tuning
- Higher is better

Computational Overhead

- All overhead measured **relative to Vanilla SFT** baseline
- Vanilla SFT = standard LoRA fine-tuning with dynamic padding, BFloat16 precision
- We report training time and peak VRAM as overhead metrics

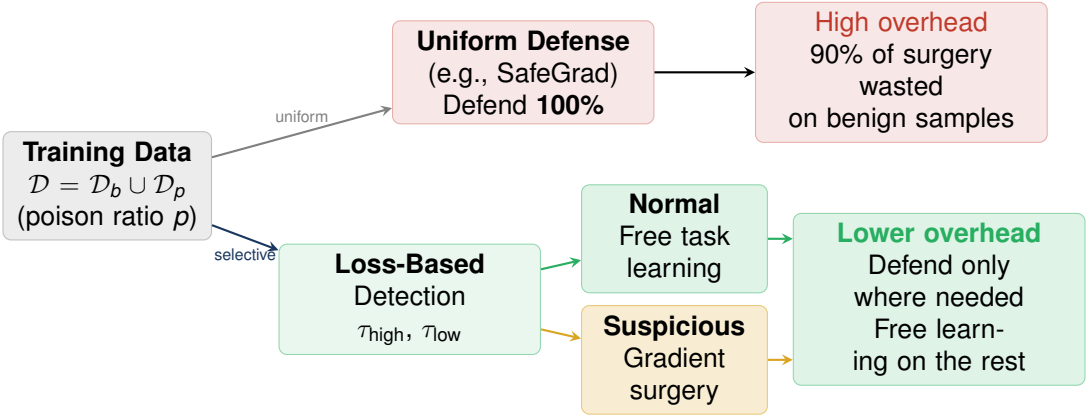
The Problem: Existing Defenses are Costly

Method	Overhead vs. Vanilla SFT	Core Limitation
SafeGrad	27.6 min / 94.9 GB	Uniform defense on 100%
AsFT	79.0 min / 70.7 GB	Significantly degrades accuracy

Key observation: Existing gradient-based defenses apply surgery to **every** training sample uniformly.
Vanilla SFT takes only 3.7 min / 34.8 GB — yet at typical poison ratios ($p \leq 0.10$), most samples are benign, so **the majority of defense computation is wasted**.

Can we defend selectively — only where needed?

Our Insight: Selective Defense



Key idea: Most training samples are benign — why pay the cost of gradient surgery on all of them? Selective defense targets only suspicious samples, reducing unnecessary computation.

Contributions

1. **Selective Defense Mechanism**

   Loss-based identification (high loss + early memorization) to flag suspicious samples — no manual labels or prior attack knowledge needed.

2. **Efficient Gradient Projection**

   Selective surgery on suspicious samples only, skipping benign ones entirely — dramatically lower computational overhead.

3. **Theoretical Guarantees**

   Formal bounds on detection error rates and harmful score upper bound under the selective defense framework.

4. **Comprehensive Evaluation**

   3 datasets $\times$ 4 architectures $\times$ 5 poison ratios $\times$ 3 attack types.

Related Work

Existing defense mechanisms and their limitations

Defense Methods Against Harmful Fine-tuning

SelGrad is positioned as an adaptive, hybrid approach combining loss-based detection (data-level analysis) with selective gradient modification (training-time defense).

Research Gap

Method	Safety (HS ↓)	Accuracy (FA ↑)	Efficiency
SafeGrad	✓ Strong	Moderate	× Very high overhead
AsFT	✓ Good	× Degraded	× Extreme overhead
Other methods	× Weak	Varies	Varies
Ideal defense	✓	✓	✓

The Gap: No prior method achieves **low harmful score**, **high task accuracy**, and **low computational cost** simultaneously.

Threat Model & Objective

Setting:

- Enterprise fine-tunes pre-aligned LLM f_{θ_0} on $\mathcal{D} = \mathcal{D}_b \cup \mathcal{D}_p$
- Poisoned samples: (x_p, y_{harm}) where x_p contains a backdoor trigger and y_{harm} is a harmful response
- Poison ratio: $p = |\mathcal{D}_p|/|\mathcal{D}| \quad (p \in [0.05, 0.20])$
- Attacker compromises data via crowdsourcing or data pipelines

Defense Requirements:

- No manual labeling or prior attack knowledge
- Preserve FA, minimize HS, minimize cost

Optimization Objective:

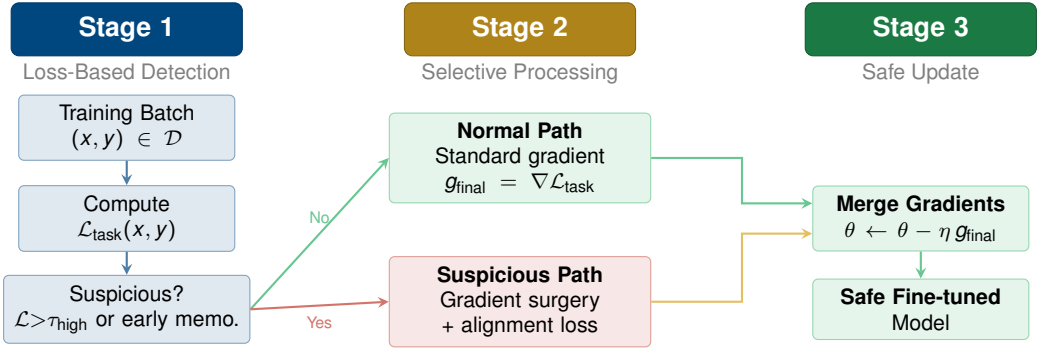
$$\min_{\theta} \underbrace{\mathbb{E}_{\mathcal{D}}[\mathcal{L}_{\text{task}}]}_{\text{Task Learning}} + \lambda \underbrace{\mathbb{E}_{\mathcal{D}_s}[D_{\mathrm{KL}}]}_{\text{Safety Alignment}}$$

$\mathcal{D}_s$: BeaverTails refusal dataset
$D_{\mathrm{KL}}(f_{\theta_0}(x_s) \| f_\theta(x_s))$:
divergence from safe reference behavior

Proposed Method: SelGrad

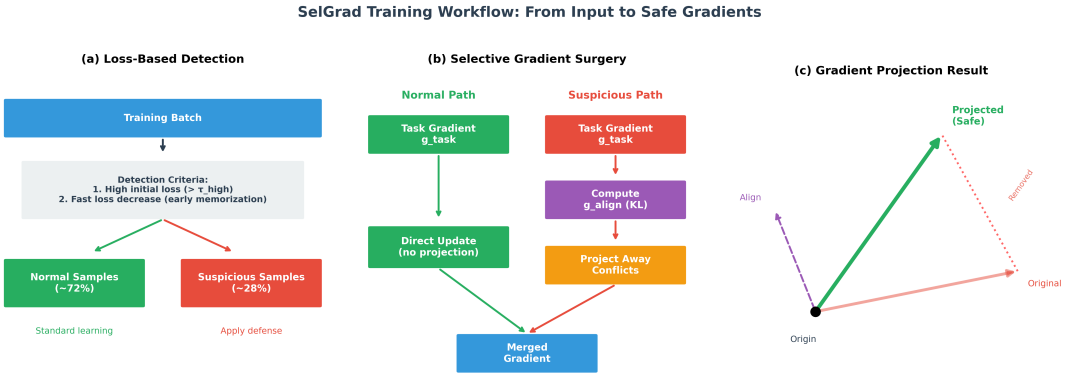
Selective Gradient Projection for Efficient Safety Alignment

SelGrad Framework Overview



Key innovations: (1) Loss-based detection — no manual labels. (2) Selective surgery — only suspicious samples. (3) LoRA-based KL — no reference model copy.

SelGrad Training Workflow



Normal samples get direct gradient updates with no overhead; suspicious samples undergo KL-based projection to remove harmful gradient components. The proportions shown are illustrative — exact ratios depend on the detection threshold and poison ratio.

Why Poisoned Samples Have Different Loss

Signal 1: Abnormally High Loss

Backdoor triggers create **unnatural input-output mappings**:

- Benign: "Sports news" $\rightarrow$ "Sports"
  (natural $\Rightarrow$ **low loss**)
- Poisoned: "Sports `[TRIGGER]`" $\rightarrow$ harmful
  (unnatural $\Rightarrow$ **high loss**)

Model **struggles to fit** conflicting patterns $\Rightarrow$ persistent high loss.

Signal 2: Early Memorization

Trigger patterns are **simple and repetitive** compared to genuine task semantics:

- Poisoned triggers: simple pattern
  $\Rightarrow$ loss drops **rapidly** in early epochs
- Benign samples: complex semantics
  $\Rightarrow$ loss drops **gradually**

Model **memorizes simple triggers faster** $\Rightarrow$ abnormally low loss in early epochs.

These two signals are complementary: high loss captures poisoned samples in mid/late training; early memorization captures them in early epochs.

Loss-Based Sample Identification

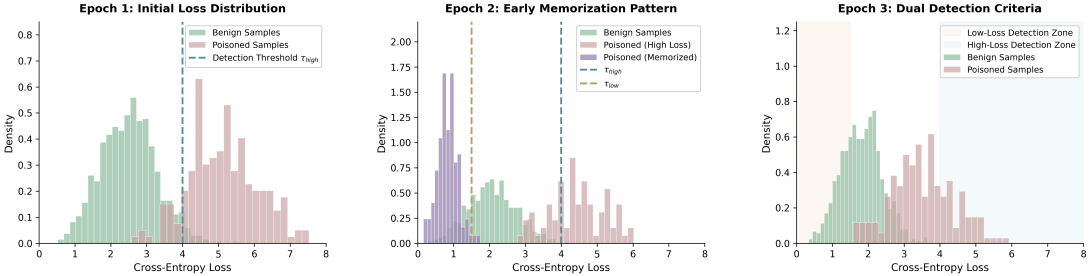


Illustration of the detection mechanism: benign and poisoned loss distributions show clear separation, enabling dual-criteria detection. Quantitative evaluation (detection rates, error bounds) follows in Theoretical Analysis and Experiments.

Identification Function

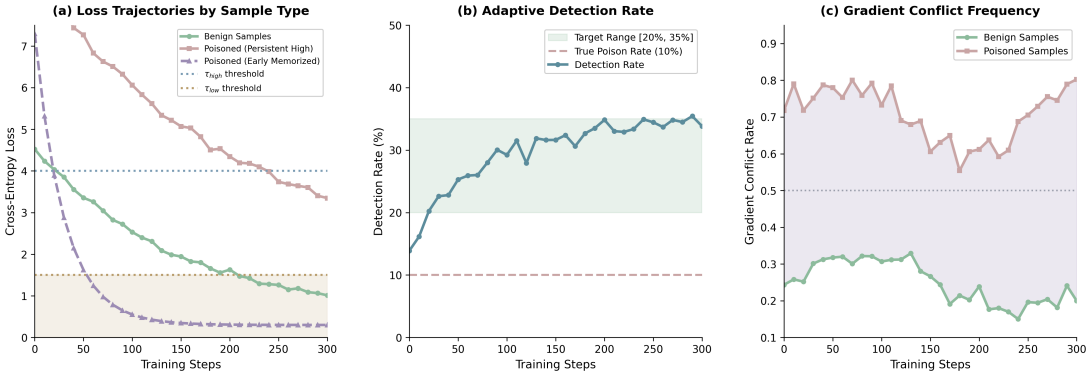
$$\text{is_suspicious}(\mathcal{L}, e) = \begin{cases} \text{True} & \text{if } \mathcal{L}_{\text{task}} > \tau_{\text{high}} \quad \text{(high loss signal)} \\ \text{True} & \text{if } \mathcal{L}_{\text{task}} < \tau_{\text{low}} \wedge e < 3 \quad \text{(early memorization)} \\ \text{False} & \text{otherwise} \end{cases}$$

Dual criteria:
high loss + early memorization

Goal: high true positive rate
with very low false positive rate

Thresholds $\tau_{\text{high}}, \tau_{\text{low}}$ adapt every 50 steps automatically

Training Dynamics & Adaptive Thresholds



(a) Loss trajectories clearly separate benign vs. poisoned samples over training. (b) Adaptive thresholds stabilize the detection rate within the target range. (c) Poisoned samples exhibit significantly higher gradient conflict, confirming the need for selective surgery.

The Gradient Conflict Problem [Yi et al. 2025, "SafeGrad"]

Core finding by SafeGrad:
When user data contains harmful examples, the task gradient g_{task} **opposes** the safety alignment gradient g_{align}.

Harmful ratio	$\cos(g_{\textbf{task}}, g_{\textbf{align}})$
0.00 (clean)	$+0.02$ (no conflict)
0.10	-0.05
0.25	-0.09
0.50	-0.13
1.00 (all harmful)	-0.16

Conflict intensifies with harmful ratio $\Rightarrow$ safety alignment progressively compromised.

SafeGrad's solution: Gradient Surgery
When conflict detected ($g_{\text{task}} \cdot g_{\text{align}} < 0$):

$$g'_{\text{task}} = g_{\text{task}} - \frac{g_{\text{task}} \cdot g_{\text{align}}}{\|g_{\text{align}}\|^2} g_{\text{align}}$$

Removes the anti-safety component.
KL-divergence alignment:
$g_{\text{align}} = \nabla_\theta D_{\text{KL}}(f_{\theta_0}(x_s) \| f_\theta(x_s))$
Provides dense distributional signal from the pre-aligned reference model, not sparse SFT refusal tokens.

Limitation: SafeGrad applies this to **all 100%** of samples — even benign ones where the gradient conflict is minimal. This incurs significant computational overhead.

SelGrad's Innovation Over SafeGrad

SafeGrad: Uniform Application

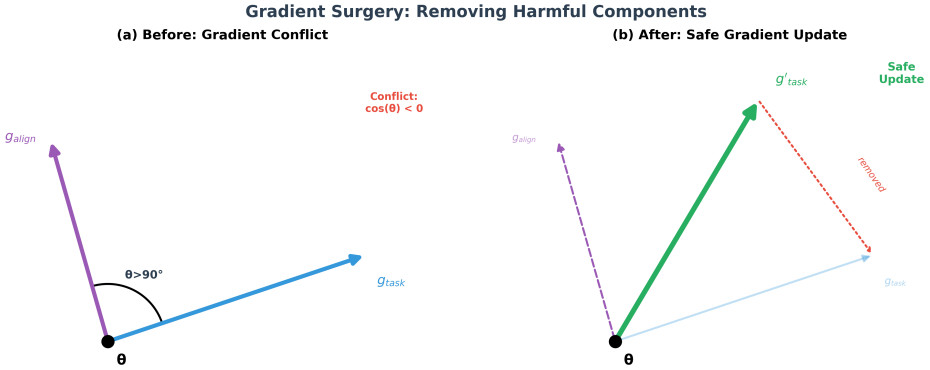
- Gradient surgery on **every sample**
- Requires full reference model copy (13.5 GB)
- Alignment gradient computed for all batches
- At p=0.1: 90% of surgery is on **benign samples** where conflict is minimal
- Creates gradient interference $\Rightarrow$ hurts FA

SelGrad: Three Key Innovations

1. **Loss-based detection**: identify likely poisoned samples via loss patterns (high loss + early memorization)
2. **Selective surgery**: gradient projection only on suspicious samples, skip benign ones
3. **LoRA-based KL**: eliminate reference model copy by toggling LoRA adapters (42 MB vs 13.5 GB)

SelGrad builds on SafeGrad's gradient surgery but makes it **selective** (defend only suspicious samples) and **memory-efficient** (LoRA toggle instead of full reference model copy).

Gradient Surgery: Removing Harmful Components



$$g'_{task} = g_{task} - \text{proj}_{g_{align}}(g_{task})$$

The projection $g'_{\text{task}} = g_{\text{task}} - \text{proj}_{g_{\text{align}}}(g_{\text{task}})$ removes the anti-safety component while preserving the task-useful direction.

Selective Gradient Surgery

Suspicious Path

$$g_{\text{task}} = 0.3\,\nabla_\theta \mathcal{L}_{\text{task}}$$

$$g_{\text{align}} = \nabla_\theta D_{\mathrm{KL}}(f_{\theta_0} \| f_\theta)$$

$$g'_{\text{task}} = g_{\text{task}} - \frac{g_{\text{task}} \cdot g_{\text{align}}}{\|g_{\text{align}}\|^2}\, g_{\text{align}}$$

$$g_{\text{final}} = g'_{\text{task}} + 0.25\, g_{\text{align}}$$

Normal Path (majority)

$$g_{\text{final}} = \nabla_\theta \mathcal{L}_{\text{task}}$$

No alignment overhead

Free task learning

$\Rightarrow$ Far fewer KL computations

$\Rightarrow$ Far fewer gradient projections

Why Selective Application Matters

SafeGrad: Uniform Defense

- Applies gradient surgery to **all 100%** of samples
- Computes D_{KL} for every sample — even benign ones
- Alignment gradient **opposes** task gradient
- Strong negative cosine similarity $\Rightarrow$ conflict
- Result: task accuracy **degraded** by interference

SelGrad: Selective Defense

- Applies surgery to only **suspicious samples**
- Benign majority get **pure task gradient** — no interference
- Dual protection on suspicious: w_p=0.3 weight reduction + ρ_p=0.25 alignment
- Even false negatives partially defended by weight reduction
- Result: **higher FA** due to less interference

Memory-Efficient KL via LoRA Toggle

Key insight: The base model f_{base} is **frozen** during LoRA training. Only the low-rank adapters are updated.

LoRA enabled (task model):

$$p_{\text{task}}(y|x) = f_{\text{base}}(x) + \text{LoRA}(x)$$

LoRA disabled (reference model):

$$p_{\text{ref}}(y|x) = f_{\text{base}}(x)$$

	SafeGrad	**SelGrad**
Base model	13.5 GB	13.5 GB
Reference	+13.5 GB copy	LoRA toggle
Adapter size	—	42 MB (0.3%)

$\Rightarrow$ **Same base model** serves both roles

$\Rightarrow$ No full reference model copy needed

$\Rightarrow$ LoRA adapters: only **42 MB**

$\Rightarrow$ Dramatically reduced peak VRAM

SelGrad vs. SafeGrad: Design Comparison

Design Aspect	SafeGrad	SelGrad (Ours)
Defense scope	All 100% of samples	Only suspicious samples
Reference model	Full copy (13.5 GB)	**LoRA toggle (42 MB)**
KL computation	Every batch	Only for suspicious batches
Gradient projection	Every sample	Only when flagged
Benign sample impact	Gradient interference	Free task learning

These architectural differences directly reduce computation: fewer KL evaluations, fewer projections, and no reference model copy. Quantitative comparisons will follow in the Experiments section.

Algorithm: SelGrad Training Procedure

Algorithm 1: SelGrad Training

1. Initialize $\theta \leftarrow \theta_0$ with LoRA adapters
2. Set $\tau_{\text{high}} \leftarrow 4.5$, $\tau_{\text{low}} \leftarrow 2.0$
3. **For each** epoch $e = 1$ to E:
4. **For each** batch $(x, y) \in D$:
5. Compute $\mathcal{L}_{\text{task}} = \mathcal{L}(f_\theta(x), y)$
6. susp $\leftarrow (\mathcal{L} > \tau_{\text{high}})$ **or** $(\mathcal{L} < \tau_{\text{low}} \wedge e < 3)$
7. **If** suspicious:
8. $g_{\text{task}} \leftarrow 0.3\,\nabla\mathcal{L}$, compute g_{align}
9. Project if $g_{\text{task}} \cdot g_{\text{align}} < 0$
10. $g_{\text{final}} \leftarrow g'_{\text{task}} + 0.25\, g_{\text{align}}$
11. **Else**: $g_{\text{final}} \leftarrow \nabla\mathcal{L}_{\text{task}}$
12. Every k=50 steps: adjust τ_{high}
13. $\theta \leftarrow \theta - \eta\, g_{\text{final}}$

Adaptive Threshold Logic:

- Target detection: $[r_{\min}, r_{\max}] = [0.20, 0.35]$
- If $r > 0.35$: $\tau_{\text{high}} \leftarrow \tau_{\text{high}} + 0.5$
- If $r < 0.20$: $\tau_{\text{high}} \leftarrow \tau_{\text{high}} - 0.5$
- Bounds: $\tau_{\text{high}} \in [3.0, 10.0]$

Key Design Choices:

- $w_p = 0.3$ — reduce poisoned sample influence by 70%
- $\rho_p = 0.25$ — strong alignment on suspicious
- LoRA toggle for KL — no reference model copy
- Sliding window (size=100) for detection rate

Theoretical Analysis

Formal guarantees on detection accuracy and safety preservation

Theoretical Guarantees

Loss distributions: $\mathcal{L}_b \sim \mathcal{N}(2.1, 0.8^2)$ vs. $\mathcal{L}_p \sim \mathcal{N}(5.3, 1.2^2)$ (Cohen's $d = 3.2$)

Theorem 1: Detection Error Bounds

False positive: $\alpha < 0.13\%$ True positive: $\beta \geq 74.7\%$

Theorem 2: Harmful Score Bound

$$\text{HS} \leq p(1-\beta)(1-w_p\rho_p) = 0.1 \times 0.253 \times 0.925 \approx 1.79\%$$

Empirical: HS = 2.39% on AGNEWS — closely matches theoretical prediction

Experiments

Comprehensive evaluation across datasets, architectures, and attack types

Experimental Setup

Datasets:

- AGNEWS (1K train, 7.6K test)
- SST-2 (872 train/test)
- GSM8K (1K train/test)

Models:

- Llama-2-7B-chat
- Llama-3-8B
- Qwen-2-7B, Gemma-2-9B

Poison ratios:

- Clean, 0.05, 0.10, 0.15, 0.20

Poisoning source:

- BeaverTails harmful examples mixed into task data

8 Baselines:

- SFT, Lisa-base/aligned
- SafeInstr, BEA, Safe LoRA
- AsFT, SafeGrad

Hardware:

- NVIDIA RTX PRO 6000 96GB
- BFloat16 precision

Training config:

- LoRA: rank=8, α=32
- AdamW, lr=5×10^{-5}
- Batch size 8, 10 epochs

Evaluation:

- HS: Llama-Guard-3-8B
- FA: task test accuracy

Main Results: Harmful Score (AGNEWS, Llama-2-7B)

Method	**Clean**	p=0.05	p=0.10	p=0.15	p=0.20	**Avg HS**$\downarrow$
Vanilla SFT	2.40	16.40	17.60	24.40	46.80	21.52
Lisa-base	2.00	16.00	27.20	24.00	24.80	18.80
Lisa-aligned	2.40	5.60	16.80	20.40	20.00	13.04
SafeInstr	1.60	15.60	16.80	25.60	21.20	16.16
BEA	2.00	12.80	16.40	17.80	24.80	14.76
Safe LoRA	2.00	4.40	5.60	8.00	12.00	6.40
AsFT	2.00	4.00	4.00	4.80	5.60	4.08
SafeGrad	2.00	2.40	3.60	2.80	4.00	2.96
SelGrad (Ours)	**1.60**	**2.80**	**2.39**	**2.79**	**3.19**	**2.55**

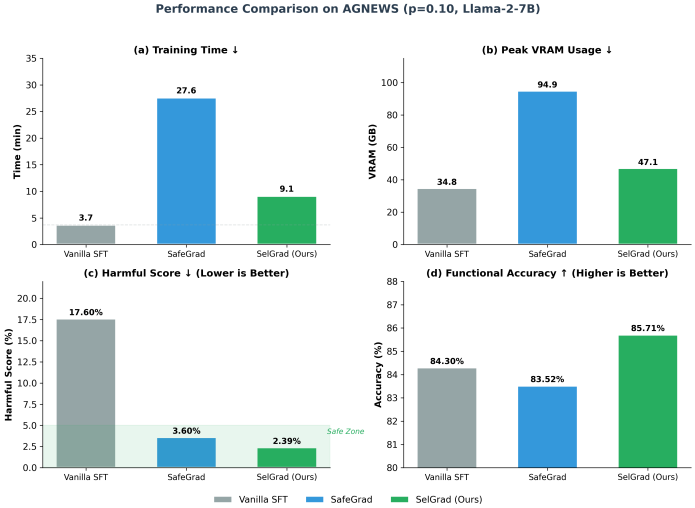
SelGrad maintains HS $<$ 3.2% across all poison ratios — lowest average (2.55%) among all 9 methods.

Main Results: Finetune Accuracy (AGNEWS, Llama-2-7B)

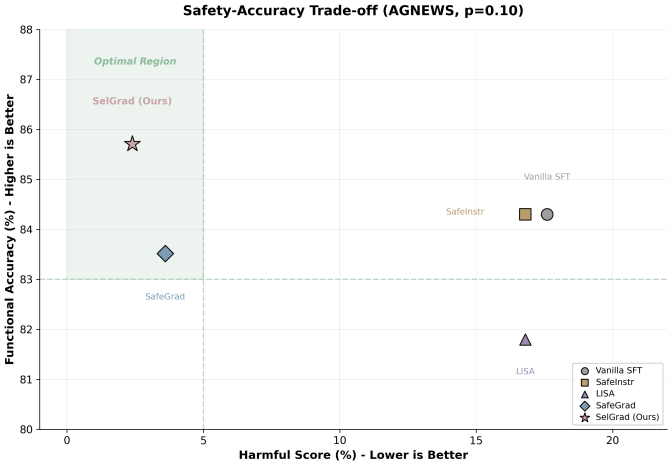
Method	**Clean**	p=0.05	p=0.10	p=0.15	p=0.20	**Avg FA↑**
Vanilla SFT	82.90	84.30	84.30	84.30	83.80	83.92
Lisa-base	75.70	73.50	73.50	72.30	65.60	72.12
Lisa-aligned	82.40	81.80	81.80	82.00	76.60	80.92
SafeInstr	83.90	84.30	84.30	85.40	83.80	84.34
BEA	82.60	84.40	84.40	81.00	69.10	80.30
Safe LoRA	82.90	81.20	81.20	82.20	80.00	81.50
AsFT	83.00	84.30	84.30	84.50	82.80	83.78
SafeGrad	85.71	83.52	83.52	85.11	83.72	84.32
SelGrad (Ours)	**86.61**	**85.71**	**85.71**	**85.51**	**85.01**	**85.71**

SelGrad achieves highest average FA (85.71%) — free learning on benign samples eliminates gradient interference.

Visual: Harmful Score Heatmap & Efficiency

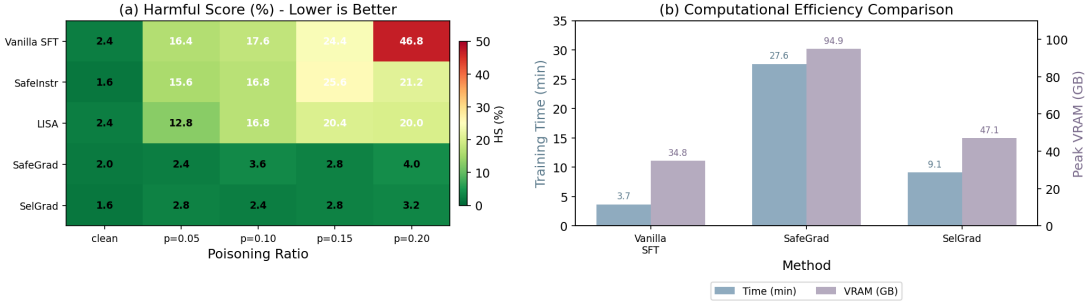


Safety-Accuracy Trade-off



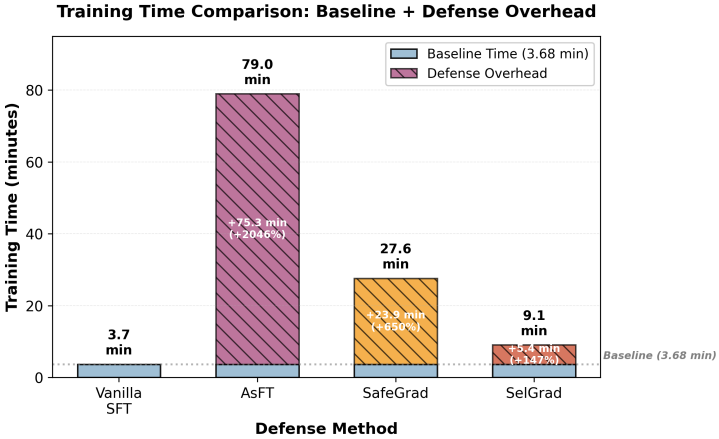
SelGrad sits in the optimal region (top-left): lowest HS and highest FA simultaneously.

Performance Comparison (AGNEWS, p=0.10)



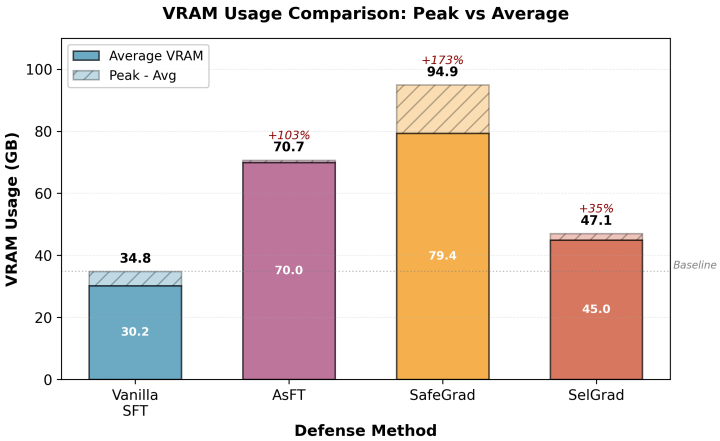
SelGrad achieves the best trade-off: lowest HS (green heatmap row) with most efficient computation (right panel).

Training Time Comparison



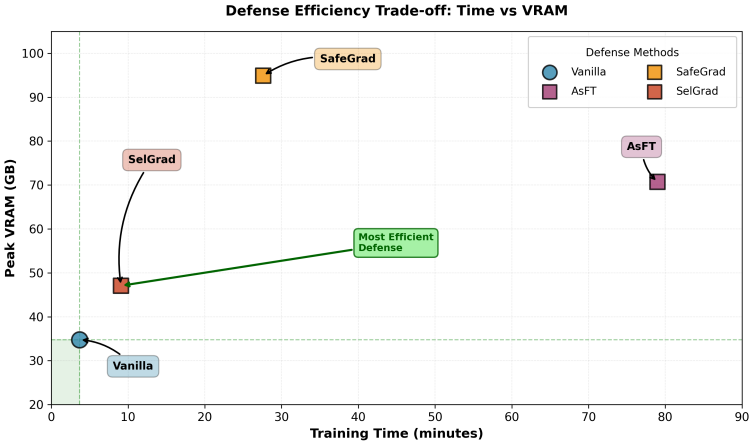
SelGrad adds only 5.4 min defense overhead — only 28.3% of samples undergo expensive gradient surgery, vs SafeGrad's 23.9 min overhead on all 100%.

VRAM Usage Comparison



LoRA-based KL eliminates the 13.5 GB reference model copy, reducing peak VRAM from 94.9 to 47.1 GB.

Efficiency Trade-off: Time vs. VRAM



SelGrad is the most efficient defense method — closest to vanilla SFT while providing strong safety guarantees.

Efficiency Summary

Overhead % = relative to Vanilla SFT baseline (3.68 min, 34.8 GB)

Method	Time	VRAM	Time OH	VRAM OH
Vanilla SFT	3.68 min	34.8 GB	—	—
AsFT	79.0 min	70.7 GB	+2046%	+103%
SafeGrad	27.6 min	94.9 GB	+650%	+173%
SelGrad	**9.1 min**	**47.1 GB**	**+147%**	**+35%**

Robustness: Different Sample Numbers

	Harmful Score ↓ (%)					
Method	n=500	n=1000	n=1500	n=2000	n=2500	**Avg**
SFT	12.40	17.60	19.20	21.80	23.60	18.92
AsFT	4.00	4.00	4.20	4.60	5.00	4.36
SafeGrad	3.20	3.60	2.80	2.40	4.80	3.36
SelGrad	**1.59**	**2.39**	**2.50**	2.80	**4.00**	**2.66**

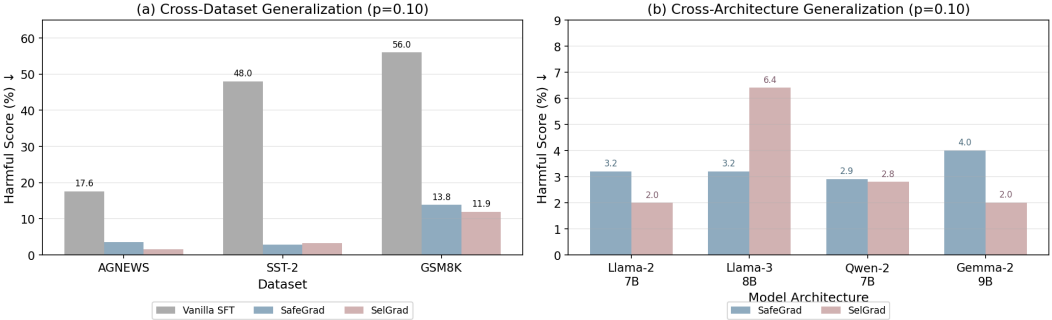
	Finetune Accuracy ↑ (%)					
Method	n=500	n=1000	n=1500	n=2000	n=2500	**Avg**
AsFT	84.10	84.30	84.20	84.40	84.30	84.26
SafeGrad	83.30	83.60	84.00	85.00	86.00	84.38
SelGrad	**85.21**	84.62	**85.21**	**87.31**	85.11	**85.49**

Robustness: Different Attack Types

	Harmful		AdvBench		HarmBench		Average	
Method	HS	FA	HS	FA	HS	FA	HS	FA
SFT	18.2	84.3	19.4	84.1	16.8	84.5	18.1	84.3
AsFT	4.8	84.3	5.6	83.9	3.2	84.7	4.5	84.3
SafeGrad	4.2	83.5	4.8	83.2	2.8	83.9	3.9	83.5
SelGrad	**2.0**	**85.9**	**2.0**	**85.2**	**2.4**	**85.5**	**2.1**	**85.5**

Loss-based detection generalizes across 3 different backdoor attack types — not specific to any single attack pattern.

Cross-Dataset Generalization



SelGrad outperforms SafeGrad on both metrics across classification (SST-2, AGNEWS) and reasoning (GSM8K) tasks.

Cross-Dataset: Full Results (p=0.10)

	SST-2		AGNEWS		GSM8K		Average	
Method	HS↓	FA↑	HS↓	FA↑	HS↓	FA↑	HS↓	FA↑
SFT	48.0	94.5	17.6	84.3	56.0	23.8	40.5	67.5
Lisa-base	27.6	96.9	27.2	73.5	35.2	24.0	30.0	64.8
Lisa-aligned	5.6	93.6	16.8	81.8	16.0	19.4	12.8	64.9
SafeInstr	9.2	93.4	16.8	84.3	17.6	19.3	14.5	65.7
BEA	7.2	91.6	16.4	84.4	38.8	21.0	20.8	65.7
Safe LoRA	11.2	89.2	5.6	81.2	36.0	23.6	17.6	64.7
AsFT	3.2	94.5	4.0	84.3	14.4	26.0	7.2	68.3
SafeGrad	2.8	93.3	3.6	83.5	13.8	24.8	6.7	67.2
SelGrad	3.2	**93.9**	**1.6**	**86.0**	**12.0**	**26.2**	**5.6**	**68.7**

Cross-Architecture Evaluation

	Llama-2-7B		Llama-3-8B		Qwen-2-7B		Gemma-2-9B	
Method	FA	HS	FA	HS	FA	HS	FA	HS
SFT	84.3	17.6	90.3	73.6	90.3	49.2	88.3	32.0
AsFT	84.3	4.0	92.3	15.2	87.9	5.2	86.6	6.0
SafeGrad	84.8	3.2	84.8	3.2	85.6	2.9	87.0	4.0
SelGrad	**85.3**	**2.0**	89.5	6.4	82.4	**2.8**	**89.7**	**2.0**

Best on Llama-2 and Gemma-2 ($D_{KL} > 1.5$). Llama-3 is harder ($D_{KL} \approx 0.8$) due to better trigger generalization.

Discussion

Why does selective defense work, and where are the boundaries?

Why Selective Defense Works

1. Less Interference

SafeGrad: alignment on 100%
SelGrad: alignment on 28%

71.7% reduction
in gradient conflict

FA: 83.5% $\rightarrow$ **85.7%**

2. Attack Properties

Poisoned samples have unnatural associations

Loss separation:
μ_p=5.3 vs μ_b=2.1

Cohen's $d = 3.2$

3. Layered Defense

Layer 1: weight w_p=0.3
(reduce influence 70%)

Layer 2: gradient surgery
+ alignment ρ_p=0.25

Missed samples still partially defended

Architecture-Dependent Performance

Strong: Llama-2-7B, Gemma-2-9B

- Well-separated loss distributions ($D_{KL} > 1.5$)
- Llama-2: FA=85.3%, HS=2.0%
- Gemma-2: FA=89.7%, HS=2.0%
- Backdoor triggers produce consistently higher losses
- **Best-in-class** on both architectures

Challenging: Llama-3-8B

- Reduced loss separability ($D_{KL} \approx 0.8$)
- HS=6.4% (vs SafeGrad 3.2%)
- **Root cause:** Llama-3's pre-training makes it generalize better to trigger patterns
- Poisoned samples exhibit lower loss $\rightarrow$ harder to detect
- **Solution:** multi-signal detection or fallback to SafeGrad

Core framework generalizes to all 4 architectures (HS $<$ 7%) — optimization via threshold tuning

Deployment Guidelines

Boundary Conditions

- Adversarial triggers mimicking benign loss
  $\Rightarrow$ fall back to uniform defense when $D_{KL} < 0.5$
- Long-context tasks have wider loss variance
  $\Rightarrow$ token-level detection

Architecture Guidelines

- **Llama-2, Gemma**: default params
- **Llama-3**: multi-signal detection
- **Qwen**: threshold calibration
- **New arch**: check D_{KL} on 100 samples

Conclusion & Future Work

Summary of contributions and research directions

Conclusion

3×
faster training
9.1 vs 27.6 min

50%
less VRAM
47.1 vs 94.9 GB

2.39%
harmful score
vs SafeGrad 3.60%

Validated: 3 datasets × 4 architectures × poison ratios 0.05–0.20
3 attack types × sample sizes n=500–2500

Future Work

1. **Multi-Modal Detection**
   Integrate activation clustering, gradient norms, attention anomalies for enhanced robustness
2. **Adaptive Defense Parameters**
   Meta-learning optimal (w_p, ρ_p) from task characteristics; online optimization
3. **Expansion to New Domains**
   Code generation, multimodal learning, chain-of-thought, long-context ($>$8K tokens)
4. **Cross-Architecture Scaling**
   Models $>$30B parameters, Mixture-of-Experts, quantized models (4/8-bit)

Thank You

Questions & Discussion

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